

Spinal Motion Restriction

Management of Patients with Potential Spinal Injuries

The long spine board **SHOULD NOT** be considered the primary mechanism for spinal motion restriction.

Patients with a serious injury should be assessed for spinal injury. Significant spinal injury typically results from high energy impacts such as motor vehicle crashes, auto-pedestrian incidents, industrial accidents, or falls. Patients with suspicion of spinal injury require immediate spinal motion restriction.

Spinal motion restriction may be accomplished using a variety of techniques. Current evidence indicates that securing a patient to the soft mattress of a stretcher using a rigid cervical collar, and webbing/safety belts should provide adequate motion restriction. Foam headblocks, extra clothing, or objects such as water bottles may be used if the patient needs lateral support. The head of the stretcher should be elevated, to further limit motion while in transport, to approximately 30 degrees as possible. The primary technique for spinal motion restriction shall consist of:

- Manual stabilization of the neck and back to prevent movement
- Placement of a rigid cervical collar, properly fitted
- Movement to the ambulance stretcher (may use backboard, scoop or slide board)
- Ambulatory patients may be allowed to sit and be gently lowered, onto stretcher mattress, while being supported thus maintaining neutral spinal alignment
- Head of the bed elevated to 30 degrees (may be lower if painful)
- Prevent movement of the head with lightweight head blocks
- Tape head and blocks in place to stretcher
- Stretcher seatbelts or webbing should be used at chest (with shoulder harness), waist and knee levels.
- When ready to transfer to next stretcher, remove tape and head blocks and seatbelts
- Move patient with manual stabilization of neck and back using slide method
- Transfer care to receiving staff members.

Patients SHOULD NOT be transported on a long spine board. Long spine backboards and scoop stretchers shall be used for extrication or movement. **The patient should be moved off the long spine board or scoop stretcher and placed on a softer and more forgiving yet sturdy surface.** Spinal motion restriction is not indicated in patients with penetrating trauma to the head, neck, or torso and no evidence of spinal injury as it will increase scene time and potentially increase bleeding.

INDICATIONS for Spinal Motion Restriction include:

- Fall or blunt trauma with altered level of consciousness
- Spinal pain or tenderness
- Neurologic complaint (e.g., numbness or motor weakness)
- Anatomic deformity of the spine
- High-energy mechanism of injury and any of the following:
 - Drug or alcohol intoxication
 - Inability to communicate
 - Distracting injury
 - Cervical axial load injury such as diving accident or significant fall with head first contact.

Many patients with minor trauma do not need spinal motion restriction. The risks of spinal motion restriction include: increased pain, patient agitation, increased movement, increased work of breathing, increased exposure to x-ray testing, and skin/muscle injury.

Spinal Motion Restriction is **NOT INDICATED in patients who meet **all of the following criteria:****

1. Normal level of consciousness (Glasgow Coma Score [GCS] 15)
2. Able to communicate with providers (age 2 yrs. and up)
3. No significant distracting injury such as long bone fracture or burns
4. No drug or alcohol intoxication
5. No abnormal neurologic findings or complaints
6. No spine tenderness or anatomic abnormality on exam
7. No neck pain with axial load to top of head
8. No Pain with patient initiated rotation or flexion-extension of neck

If spinal pain prevents the patient from moving to a neutral inline position, then spinal motion restriction should be performed in the position found.

If a patient requires intubation, spinal motion restriction should be maintained by a team member holding the head in neutral position throughout the intubation procedure. The cervical collar can be (re)applied to the patient post intubation.

As with all medical procedures, patients may refuse any or all parts of immobilization. Standard processes should be utilized to validate decision-making capacity and comprehension of risk and benefit to all medical procedures. Documentation of decision-making capacity and understanding of risks must be included in the PCR.

SMR Check List

INDICATIONS for Spinal Motion Restriction include:

- ☐ Fall or Blunt trauma with altered level of consciousness
- ☐ Spinal pain or tenderness
- ☐ Neurologic complaint (e.g., numbness or motor weakness)
- ☐ Anatomic deformity of the spine
- ☐ High-energy mechanism of injury and any of the following:
 - ☐ Drug or alcohol intoxication
 - ☐ Inability to communicate or participate in exam
 - ☐ Distracting injury
 - ☐ Cervical axial load injury, such as diving accident or significant fall with head first contact

Spinal Motion Restriction is **NOT INDICATED in patients who meet **all of the following criteria:****

- ☐ Normal level of consciousness (Glasgow Coma Score [GCS] 15)
- ☐ Able to communicate with providers (age 2 yrs. and up)
- ☐ No significant distracting injury such as long bone fracture or significant burns or soft tissue injuries
- ☐ No drug or alcohol intoxication
- ☐ No abnormal neurologic findings or complaints
- ☐ No spine tenderness to palpation or anatomic abnormality on exam
- ☐ No neck pain with axial load to top of head
- ☐ No complaint of midline neck pain with patient initiated rotation or flexion-extension of neck